



## Title

**lpbwsselect** — Bandwidth Selection Procedures for Local Polynomial Regression Estimation and Inference.

## Syntax

```
lpbwsselect yvar xvar [if] [in] [, eval(gridvar) neval(#) deriv(#) p(#)
kernel(kernelfn) bwselect(bwmethod) bwcheck(#) imsegrid(#) vce(vcetype
[vceopt]) bwregul(#) separator(#) interior genvars weights(wvar)
masspoints(check|off) ]
```

## Description

**lpbwsselect** implements bandwidth selectors for local polynomial regression point estimators and inference procedures developed in [Calonico, Cattaneo and Farrell \(2018\)](#). See also [Calonico, Cattaneo and Farrell \(2022\)](#) for related optimality results. It also implements other bandwidth selectors available in the literature.

A detailed introduction to this command is given in [Calonico, Cattaneo and Farrell \(2019\)](#).

Companion command is: [lprobust](#) for local polynomial point estimation and inference procedures.

Related software useful for empirical analysis is described in the following website:

<https://nppackages.github.io/>

Requires Stata 14 or later.

## Options

**eval(gridvar)** specifies the grid of evaluation points for *xvar*. By default it uses 30 equally spaced points over the support of *xvar*.

**neval(#)** specifies the number of evaluation points to estimate the regression functions. Default is 30 evaluation points.

**deriv(#)** specifies the order of the derivative of the regression functions to be estimated. Default is **deriv(0)**.

**p(#)** specifies the order of the local polynomial used to construct the point estimator. Default is **p(1)** (local linear regression).

**kernel(kernelfn)** specifies the kernel function used to construct the local-polynomial estimator(s). Options are: **triangular**, **epanechnikov**, **uniform** and **gaussian**. Default is **kernel(epanechnikov)**.

**bwselect(bwmethod)** bandwidth selection procedure to be used. Options are:  
**mse-dpi** second-generation DPI implementation of MSE-optimal bandwidth. Default choice.

**mse-rot** ROT implementation of MSE-optimal bandwidth.

**imse-dpi** second-generation DPI implementation of IMSE-optimal bandwidth.

**imse-rot** ROT implementation of IMSE-optimal bandwidth.

**ce-dpi** second generation DPI implementation of CE-optimal bandwidth.

**ce-rot** ROT implementation of CE-optimal bandwidth.

Note: MSE = Mean Square Error; IMSE = Integrated Mean Squared Error; CE = Coverage Error; DPI = Direct Plug-in; ROT = Rule-of-Thumb.

Default is **bwselect(mse-dpi)**. For details on implementation see [Calonico, Cattaneo and Farrell \(2019\)](#).

**bwcheck(#)** specifies an optional positive integer so that the selected bandwidth is enlarged to have at least # effective observations available for each evaluation point.

**imsegrid**(#) number of evaluations points used to compute the IMSE bandwidth selector. Default is 30 points.

**vce**(*vcetype* [*vceopt1*]) specifies the procedure used to compute the variance-covariance matrix estimator. Options are:

- vce**(*nn* [*nnmatch*]) heteroskedasticity-robust nearest neighbor variance estimator with *nnmatch* minimum neighbors. Default when no cluster variable is supplied.
- vce**(*hc0*) heteroskedasticity-robust plug-in residuals, no weights.
- vce**(*hc1*) heteroskedasticity-robust plug-in residuals, HC1 weights.
- vce**(*hc2*) heteroskedasticity-robust plug-in residuals, HC2 weights (leverage).
- vce**(*hc3*) heteroskedasticity-robust plug-in residuals, HC3 weights (jackknife).
- vce**(*cr1* *clustervar*) cluster-robust CR1 variance with  $((n-1)/(n-k)) * (G/(G-1))$  multiplier. Default when a cluster variable is supplied.
- vce**(*cr2* *clustervar*) cluster-robust CR2 variance (Bell-McCaffrey block-adjusted residuals).
- vce**(*cr3* *clustervar*) cluster-robust CR3 variance (block jackknife,  $(G-1)/G$  multiplier).
- vce**(*cluster* *clustervar*) alias for **vce**(*cr1* *clustervar*).

Legacy: **vce**(*hc0*/*hc1* *clustervar*) is remapped to **cr1**, **vce**(*hc2* *clustervar*) to **cr2**, **vce**(*hc3* *clustervar*) to **cr3**. **vce**(*nncluster* ...) is no longer supported and is remapped to **cr1** with a warning.

Default is **vce**(*nn* 3) when no cluster, **vce**(*cr1*) when a cluster variable is supplied.

**bwregul**(#) specifies scaling factor for the regularization term added to the denominator of the bandwidth selectors. Setting **bwregul**(0) removes the regularization term from the bandwidth selectors. Default is **bwregul**(1).

**separator**(#) draws separator line after every # variables; default is separator(5).

**interior** option to set all evaluation points to be interior points.

**genvars** generates new variables **lpbwselect\_eval**, **lpbwselect\_h**, and **lpbwselect\_b** storing the evaluation points and selected bandwidths (Or per-method bandwidths when **bwselect**(*all*)).

**weights**(*wvar*) optional vector of non-negative observation weights (multiplicative with kernel weights in all bandwidth-selection steps).

**masspoints**(*check*|*off*) how to handle evaluation points with few unique *xvar* values within bandwidth. Default is **check**; **off** disables the warning.

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#### Example: Cholesterol Trial Data

```
Setup
. use nprobust_data.dta

Second-generation DPI implementation of MSE-optimal bandwidth
. lpbwselect cholf chol1 if t==0
```

#### Stored results

**lpbwselect** stores the following in **e()**:

##### Scalars

|             |                                                                        |
|-------------|------------------------------------------------------------------------|
| <b>e(N)</b> | sample size after listwise deletion                                    |
| <b>e(p)</b> | order of the polynomial used for estimation of the regression function |
| <b>e(q)</b> | order of the polynomial used for bias correction                       |

#### Macros

|                      |                                    |
|----------------------|------------------------------------|
| <b>e(cmd)</b>        | command name                       |
| <b>e(yvar)</b>       | name of dependent variable         |
| <b>e(xvar)</b>       | name of running variable           |
| <b>e(bwselect)</b>   | bandwidth selection choice         |
| <b>e(kernel)</b>     | kernel choice                      |
| <b>e(vce_select)</b> | vce choice (post-remap, lowercase) |
| <b>e(vce_type)</b>   | vce display label                  |
| <b>e(clustvar)</b>   | cluster variable (if specified)    |

#### Matrices

|               |                   |
|---------------|-------------------|
| <b>e(bws)</b> | estimation result |
|---------------|-------------------|

#### **References**

- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2018. On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference. *Journal of the American Statistical Association*, 113(522): 767-779.
- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2019. nprobust: Nonparametric Kernel-Based Estimation and Robust Bias-Corrected Inference. *Journal of Statistical Software*, 91(8): 1-33. doi: [10.18637/jss.v091.i08](https://doi.org/10.18637/jss.v091.i08).
- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2022. Coverage Error Optimal Confidence Intervals for Local Polynomial Regression. *Bernoulli*, 28(4): 2998-3022.

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